Course: **GRC103 — Risk Assessment and Management Techniques**
Assessment: **Advanced Cybersecurity Risk Quantification and Control Investment Analysis**
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Phase 1: Basic Metric & Efficiency Calculations

Step 1: Fundamentals Rates

Campaign	Email Open Rate	Click-Through Rate	Credential Submission Rate	Data Entry Rate
A	(82/100)×100 = **82%**	(47/100)×100 = **47%**	(18/100)×100 = **18%**	(5/100)×100 = **5%**
B	(200/250)×100 = **80%**	(150/250)×100 = **60%**	(75/250)×100 = **30%**	(25/250)×100 = **10%**
C	(72/80)×100 = **90%**	(60/80)×100 = **75%**	(12/80)×100 = **15%**	(8/80)×100 = **10%**

Step 2: Efficiency Metrics

Conversion Rate from Open to Click:

Campaign A: (47/82)×100 = **57.3%**
Campaign B: (150/200)×100 = **75%**
Campaign C: (60/72)×100 = **83.3%**

Post-Click Credential Submission Rate:
Campaign A: (18/47)×100 = **38.3%**
Campaign B: (75/150)×100 = **50%**
Campaign C: (12/60)×100 = **20%**

Phase 2: Advanced Risk Quantification

Step 3:Calculate Organizational Risk Exposure

Campaign	Overall Risk Score	Financial Exposure	Priority Level	Recommended Action
A	MEDIUM-HIGH	$156,600,000	2	Enhanced Training
B	**CRITICAL**	**$652,500,000**	**1**	**IMMEDIATE MFA**
C	HIGH	$104,400,000	2	Executive Training

For Campaign B (most successful):

- Expected Credentials Stolen: 2,000 × (75/250) = 600 **employees**
- Expected Data Breaches: 600 × 25% probability = 150 **breaches**

Calculate for Campaign A:

- p = 0.18, n = 100
- Margin of Error = 1.96 × √[0.18(1-0.18)/100] = 7.53%
- True Rate Range: 10.5% to 25.5%

Phase 4: Executive Reporting

To: Board of Directors, CISO, CFO
From: GRC Risk Analysis Team
Date: 1 November 2016
Subject: CRITICAL: Quantitative Phishing Risk Assessment & $652M Exposure

1. Executive Summary:
Our phishing simulation analysis reveals a **CRITICAL risk exposure of $652 million** annually. The "IT Password Reset" campaign showed a 30% credential theft rate, projecting to 600 compromised accounts organization-wide. Immediate action is required.

2. Key Quantitative Findings:

Risk Metric	Value	Industry Average	Severity
Overall Credential Theft Rate	22.1%	15.3%	HIGH
Maximum Campaign Success Rate	30%	18.7%	CRITICAL
Annualized Loss Expectancy	$652M	$285M	SEVERE
MFA Implementation ROI	724,175%	350%	EXCELLENT

3. Detailed Financial Analysis:

Current State Risk:

- Annualized Loss Expectancy: **$652,500,000**
- Probability of Major Breach: **93.75%** (150 expected breaches)
- Per-Employee Risk: **$326,250**

Proposed Security Investment Portfolio:

Control	Cost	Risk Reduction	Net Benefit	ROI	Priority
MFA Implementation	$90,000	$651.8M	$651.7M	724,175%	**1**
Enhanced Training	$100,000	$391.5M	$391.4M	391,400%	**2**
Email Filtering	$50,000	$130.5M	$130.4M	260,800%	**3**

4. Strategic Recommendations:

IMMEDIATE ACTIONS (Q1):

1. **Implement MFA enterprise-wide** - $90,000 investment
2. **Launch targeted phishing simulation program** - $25,000
3. **Conduct emergency security awareness training** - $100,000

STRATEGIC INITIATIVES (FY2024):

1. **Deploy advanced email security gateway** - $50,000
2. **Establish continuous security monitoring** - $150,000
3. **Implement security behavior analytics** - $75,000

5. Expected Risk Posture After Controls:

Metric	Current	With Controls	Reduction
Credential Theft Rate	22.1%	2.2%	90%
Annual Loss Expectancy	$652M	$65M	90%
Breach Probability	93.75%	25%	73.3%

Total Investment Required: $490,000
Total Risk Reduction: $587,500,000
Net Financial Benefit: $587,010,000

Bonus Task

1. Break-Even Calculation

Break - Even Breaches = Total Security Investment/Cost per Breach
Break - Even Breaches = $490, 000/$4, 350, 000 = 0.1126 breach

2. Executive Business Interpretation

- **Fractional Exposure:** Preventing just **0.113 of a breach** (or approximately **11.3%** of a single breach incident) fully justifies the **$490,000** investment portfolio.
- **Single Breach ROI:** Preventing just **1 single major data breach** saves GlobalTech **$4,350,000**. After deducting the **$490,000** program cost, this single prevention yields a **net financial benefit of $3,860,000** (an immediate **787.76% ROI**).
- **Bottom Line:** Because the expected baseline is **150 breach incidents** ($652.5M$ exposure), reducing the risk by even a small fraction makes the security investment an absolute no-brainer for executive leadership.